# Nisong Monyimba

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github.com/NisongMonyimba/AppliedMathThesis

## Research Interests

Applied Mathematics • Stochastic Analysis • Mean-Field Games • Stochastic Control • Scientific Computing • Numerical Analysis • Optimal Transport

## Research Profile

My background is in biomedical and manufacturing engineering, where years of applying statistical and computational methods – Monte Carlo simulation, design of experiments, statistical process control – to real quantitative problems gradually pulled my interest toward the mathematical structures underlying those methods, rather than their engineering applications alone. Since 2025 I have pursued self-directed graduate-level study in real analysis, stochastic processes, and stochastic control theory, culminating in an independent research program in mean-field games with common noise: an MSc thesis establishing a corrected local well-posedness theory for the associated stochastic maximum principle, and two companion preprints developing quantitative contraction estimates and a numerically verified particle/finite-difference scheme. My research goal is to work at the intersection of stochastic analysis, control theory, and scientific computing – particularly the numerical analysis of mean-field and McKean–Vlasov systems – and I am applying to PhD programs in applied mathematics for Fall 2027 to pursue this work under formal doctoral training.

## Research Outputs

**Quantitative Contraction Estimates for Common-Noise Mean-Field Games** 2026

- Establishes a local contraction estimate, on an explicit time horizon, for the fixed-point map characterising equilibria of mean-field games with common noise; extended to arbitrary horizons via a standard continuation argument.
- PDF | Source

**Numerical Approximation of Common-Noise MFG via Particle Methods & Finite Differences** 2026

- Combined Euler–Maruyama, particle-method, and finite-difference scheme for the coupled system characterising common-noise mean-field game equilibria; every reported convergence rate is the direct, reproducible output of executed code.
- PDF | Source & reproducible code

**Quantitative Convergence Analysis of Stochastic Maximum Principles for Mean-Field Games with Common Noise** 2026

- PDF | Full repository (code, tests, Lean 4 formal verification)

## Education

**Biomedical Engineering, M.Sc. — Arizona State University** Graduated May 2022

*Ira A. Fulton Schools of Engineering; MasterCard Foundation Scholar (Accelerated Master's, 3+1+1); Dean's List: Fall 2020, Spring 2021*

**Biomedical Engineering, B.Sc. — KNUST** Graduated May 2021

*Kwame Nkrumah University of Science & Technology; MasterCard Foundation Scholar; First Class Honours*

**Relevant quantitative coursework:** Algebra, Calculus with Analysis, Calculus with Several Variables, Differential Equations, Numerical Methods, Probability and Statistics, Finite Element Modeling, C Programming, Object-Oriented Programming, Signals and Systems, Biosignal Processes and Analysis, Embedded Systems.

Since 2025, self-directed graduate-level study (independent of formal coursework) in real analysis, stochastic processes, and stochastic control theory, undertaken in connection with an independent research program in mean-field games with common noise (see Research Statement and supplementary repository).

## Master's Applied Project & Undergraduate Capstone Project

**Master's Applied Project: Trophoblast-on-Chip Microfluidic Device** 2022

- Investigated trophoblast growth to quantify proliferation dynamics (cell counts over time, growth curves, proliferation index) using Fiji to extract quantitative growth metrics from laboratory cell images.
- Performed statistical analysis including growth-curve comparison across conditions (two-way ANOVA), doubling-time comparison between treatments (unpaired t-test), and proliferation-marker quantification (Kruskal–Wallis).

- Developed microfluidic channel designs in SolidWorks, simulated laminar flow and shear profiles in COMSOL Multiphysics, and validated models against analytical equations.

**Capstone Project: Novel ECMO Cannula Design, Arizona State University** 2021

- Designed and simulated ECMO cannula flow in COMSOL Multiphysics, optimizing shear stress and velocity distribution; built a 3D model in SolidWorks and simulated flow in ANSYS Fluent.
- Quantified flow uniformity and reduced clot-formation risk by optimizing inlet geometry and flow-rate parameters.
- Performed finite element analyses (FEA) in ANSYS and SolidWorks to assess performance and safety margins under physiological load conditions.

## Research & Lab Work

**Research Assistant, Weaver Lab, Arizona State University** 2021–2022

- Analyzed trophoblast proliferation dynamics by quantifying cell counts over time and fitting growth curves.
- Used Fiji to extract quantitative metrics from laboratory images, including colony expansion area and fluorescence intensity, supporting rigorous modeling of cellular growth patterns.
- Quantified cell proliferation rates from fluorescence microscopy data using Fiji and GraphPad Prism; developed statistical plots and regression analyses to characterize proliferation and viability.
- Developed Python-based probabilistic simulations and Monte Carlo analyses to evaluate variability in material properties and stochastic loading, improving predictive accuracy.

**Research Assistant, Dr. Pizziconi's Lab, Arizona State University** 2022–2023

- Designed and optimized bone fixation screws using Halpin–Tsai micromechanics models to predict composite behavior under mechanical load.
- Created and refined 3D CAD models in SolidWorks, ensuring dimensional accuracy, manufacturability, and integration with experimental validation workflows.
- Trained and mentored lab members in modeling, simulation, and experimental validation, connecting theoretical mathematics with engineering practice.
- Collaborated with interdisciplinary teams to iterate designs from simulation results; documented and presented research outcomes supporting publication efforts and potential clinical translation.

**Research Assistant, Medical Device Regulation (MDR) Project, KNUST** 2020–2021

- Analyzed regulatory frameworks governing medical devices across 20 African countries, identifying disparities in approval processes, safety standards, and market accessibility.
- Used publicly available regulatory data to categorize countries by regulatory maturity, from fully established authorities to emerging or informal frameworks.

## Internship Experience

**REU Internship, Case Western Reserve University** 2020

- Studied biological design and product development principles; designed Arduino-based air-quality sensors and prototyped PCB systems using EAGLE CAD.
- Collaborated with interdisciplinary teams to apply quantitative modeling to life-science problems.

**BME Internship, Komfo Anokye Teaching Hospital, Kumasi, Ghana** 2019

- Supported medical equipment maintenance and calibration; performed voltage and current testing.
- Documented device performance data and proposed engineering solutions to minimize error variance; presented findings to senior biomedical engineers and clinicians.

**Engineering Internship, Vodafone (Telecom) Headquarters, Accra, Ghana**

- Designed and proposed a telemedicine smartphone application concept for patient monitoring.
- Assisted in vendor management and quality audits for customer-facing telecom hardware systems.

## Professional Experience

**Senior Manufacturing Engineer, Abbott Laboratories** 2024–2025

- Optimized medical device grinding precision by 31.7% through DOE, ANOVA, and SPC-driven process improvements.
- Implemented Python-based control dashboards for real-time statistical quality monitoring.
- Applied Monte Carlo simulations, PCA, and variance component analysis to identify and mitigate process variability.
- Authored and executed FDA- and ISO 13485-compliant validation protocols.

**Quality Engineer, Vastek Inc., San Diego, CA** 2023–2024

- Designed devices and implemented engineering design practices targeting efficiency improvements; worked with a cross-functional team of medical professionals and engineers.
- Performed design analyses, requirements reviews, and revisions; wrote technical reports and protocols.

- Conducted FMEA, Bayesian risk analysis, and fault-tree analysis to quantify cascading failure probabilities.

**Med Tech R&D Engineer, Mayo Clinic & Arizona State University** 2022–2023

- Designed bone fixation screws using Halpin–Tsai equations and Mori–Tanaka models.
- Ran nonlinear FEA simulations to analyze fatigue, stress distribution, and deformation behavior.

## Teaching Experience

**Courses Prepared to Teach**

*Based on completed undergraduate coursework, self-directed graduate-level study in real analysis and stochastic processes, and the teaching and tutoring experience below*

- Calculus I & II, Linear Algebra, Differential Equations, Numerical Methods, Probability & Statistics, Engineering Mathematics

**Teaching Assistant, KNUST** 2019

- Tutored biomedical engineering undergraduates in calculus, electrical circuits, and biostatistics; assisted with lab supervision, grading, and practical demonstrations.

**Science & Math Teacher, NSMQ Team, Kwanyako Senior High School** 2016–2018

- Prepared and led students for regional and national science and math competitions, emphasizing teamwork, critical thinking, and problem-solving.

**Math Teacher, Witness Academy** 2016–2017

- Taught core mathematics and physics; prepared junior-high students for final exams (BECE).

## Volunteering, Community Service & Projects

- **Group Homes Volunteer (Autism Support):** assisted residents with learning and recreational activities.
- **MasterCard Foundation Community Projects:** led sanitation, clean-water, and youth education projects in Ghana and Arizona.
- **Oforikrom Outreach Project:** organized public school clean-ups, donations, and science tutoring.
- **Vocational Training:** trained four youths in electrical wiring and biofill installation.
- **Personal Tutoring Initiative:** mentored underprivileged students in mathematics; three students gained high-school admission and two gained college admission, all first-generation in their families.
- Organized community cleaning (gutters, weeds, trash) in Ashaiman-Zenu to reduce contamination and malaria risk.

## Awards & Competitions

- MasterCard Foundation Scholarship, Arizona State University and KNUST
- MORE Fellowship Award, Arizona State University
- Vodafone Telecommunications Design Competition, 1st Place
- Best Student, WASSCE (2016), Kwanyako Senior High School
- Best Student, BECE (2013), Zenu T.M.A No. 3 JHS
- 2nd Place, KNUST Engineering & Medical School Innovation Challenge
- Dean's List, Arizona State University

## Leadership & Mentorship

- Assistant Senior Prefect, Zenu T.M.A Junior High School
- Leader, NSMQ Science & Math Quiz Team, Kwanyako SHS
- Peer Mentor, MasterCard Foundation Scholars (KNUST & ASU)
- Community Project Facilitator, MasterCard Foundation Ghana
- Member, Young African Leaders Initiative (YALI)
- Club Executive, Biomedical Engineering Society (KNUST & ASU)

## Certifications

- Lean Six Sigma (Green Belt)
- Design of Experiments (Minitab & JMP)
- Global Leadership & Project Management (YALI / ASU Global Leadership Academy)

## Professional Memberships

Society of Manufacturing Engineers (SME) • National Society of Black Engineers (NSBE) • Biomedical Engineering Society (BMES) • ASU Global Leadership Academy • Toastmasters International • KNUST Peer Counsellors & MasterCard Peer Mentors • Ghana Engineering Students Association • TraTech

## Technical Skills

**Programming & Computation:** Python (NumPy, SciPy, statsmodels, pandas), MATLAB, R, Git/GitHub, SQL, Java, C, LaTeX (Overleaf).

**Statistical & Numerical Modeling:** Monte Carlo simulation, ANOVA, DOE, SPC, PCA, GraphPad Prism, Minitab, JMP, Octave/GNU.

**Engineering Simulation & CAD:** SolidWorks, ANSYS Fluent, COMSOL Multiphysics, Autodesk Fusion 360, SimScale.

**Other:** Fiji/ImageJ, BioRender, Figma, Inkscape, LabVIEW, LTSpice, Windchill, Zotero.